Global Noise Reduction versus Phonetic Preservation in Wiener Speech Enhancement

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Project website | MATLAB code and data | Video presentation

Abstract

This report examines how short-time Fourier transform (STFT) Wiener filtering affects speech sounds when overall signal-to-noise ratio (SNR) improves. Ten labelled English utterances from ten speakers were mixed with kitchen, cafeteria and traffic noise at four input SNRs. Two noise excerpts per utterance–noise pair produced 240 mixtures. The gain estimated from each mixture was applied separately to the known clean speech and noise to measure speech loss and noise reduction. Mean error SNR improved by 6.18 dB, while mean short-time objective intelligibility (STOI) decreased from 0.8432 to 0.8385. Mean speech retention was −1.24 dB for vowels, −2.53 dB for fricatives and −2.51 dB for stops and affricates. Fricatives lost a larger proportion of speech energy than vowels, although unweighted gain and high-frequency retention gave different class rankings. A paired experiment with a 16 ms window reduced the number of frames crossing phone boundaries but increased speech attenuation. Overall noise reduction therefore occurred alongside unequal speech preservation. The differences between classes depended on the metric and noise condition.

Index Terms—Wiener filtering, speech enhancement, phonetic preservation, STFT, speech distortion.

I. Introduction

A speech enhancer can reduce background noise while also removing part of the speech. A whole-sentence score combines these effects and gives little detail about individual sounds. Vowels usually contain strong harmonics, fricatives contain aperiodic energy, and stops include brief transitions. These differences in spectral structure and duration can lead to different gains when a filter uses local estimates of speech and noise power.

The research question is: Does STFT Wiener filtering preserve speech energy equally across broad phonetic classes when global error SNR improves? Speech retention and spectral measurements are used to compare the classes. These measurements describe changes in the signal; their effect on speech recognition requires a separate perceptual assessment.

The study applies a classical single-channel Wiener filter to controlled speech–noise mixtures. The known clean speech and added noise allow their contributions to the output to be measured separately. Comparisons cover ten speakers, three environmental recordings and four input SNRs. A second configuration uses a shorter analysis window to test whether reducing overlap with neighbouring phones improves speech preservation.

II. Literature Review

A. Gain estimation and speech distortion

For uncorrelated speech and noise, the Wiener gain minimises mean-square error and equals speech power divided by total power [1]. The amount of suppression therefore depends directly on the power estimates. An error in either estimate changes the gain, particularly when speech is weak or noise varies. A minimum gain limits attenuation, but the same power-based rule still applies to every speech sound.

Kim and Loizou [2] separated amplification and attenuation distortion in listening experiments. Attenuation alone had a relatively small effect on intelligibility under their test conditions. Errors of similar size can therefore have different perceptual effects, so minimising squared error need not maximise intelligibility. Their work supports examining the types of distortion separately. The component measurements used here quantify speech and noise energy, while the perceptual effects tested in [2] remain outside this experiment.

B. Phonetic structure and evaluation

Monir, Magron and Serizel [3] compared FaSNet, MVDR and Tango at phoneme scale in noise and reverberation. Plosives remained difficult, while nasals and sibilants often benefited from enhancement. Their objective measurements locate differences among sounds, whereas the listening experiments in [2] assess recognition. The multichannel systems in [3] also use spatial information that a single-channel Wiener filter cannot access. Their findings support phoneme-level evaluation, but the class ranking must be measured again for this filter and these recordings.

Kong, Mullangi and Kokkinakis [4] classified fricative groups using spectral features from 8 ms frames. Their results show that short spectral segments contain information that distinguishes these sounds. Classification accuracy measures a different outcome from speech retention, so it cannot directly evaluate the present filter. Shadle et al. [5] examined how bandwidth and analysis choices affect fricative measurements, including information above 8 kHz. This supports frequency-band analysis and identifies a limit of the present 16 kHz recordings, which cover frequencies only up to 8 kHz.

STOI predicts intelligibility from the correspondence between short-time spectral envelopes [6]. Here it gives one score per utterance and does not identify which phone caused a score change. The evaluation therefore also measures speech and noise retention within each phone and compares the clean and processed spectra. Error SNR measures the total reconstruction error. Together, these metrics show whether a better global score accompanies speech loss, spectral change or both.

C. Rationale for the experiment

A fixed Wiener configuration makes the relationship between estimated power and applied gain easy to inspect and requires no model training. Its main limitation is dependence on power estimates that may be inaccurate. Controlled mixtures provide exact speech and noise references, and the same estimated gain can be applied to both for evaluation. Supplied phone boundaries provide the timing for class comparisons without adding a classifier. The remaining question is how well a global error score describes preservation across speech classes in this baseline. The experiment measures that relationship rather than ranking enhancement systems.

III. Methods

A. Speech and noise data

The speech sample contains the first eligible utterance from each of ten distinct speakers encountered in the LibriSpeech Segment dev.clean subset [7], [8]. Eligibility required a duration of 3–8 s, all four speech classes, a silence interval, recognised symbols and continuous timing labels. The selected clips last 4.17–6.60 s and total 52.33 s. Each is mono at 16 kHz. The selection manifest records the utterance, speaker, duration and transcript. This is a small sample selected for phonetic coverage, rather than a random population sample. Table I groups its 549 labelled intervals.

TABLE I. Phone classes and segment counts.

Class	Phone symbols used	Segments
Vowels	AA AE AH AO AW AY EH ER EY IH IY OW OY UH UW	192
Nasals / approximants	M N NG L R W Y	113
Fricatives	F V TH DH S Z SH ZH HH	104
Stops / affricates	P B T D K G CH JH	97
Silence	[SIL]	43

The supplied alignments have not been manually verified. Fricatives include voiced sounds and HH; a stop label covers the complete interval, rather than a separately marked release burst. Silence is included to assess gain and residual noise, but excluded from speech-retention comparisons.

The noise recordings are DEMAND kitchen (DKITCHEN), cafeteria (PCAFETER) and traffic (STRAFFIC), using channel 1 at a sample rate of 16 kHz [9]. Two reproducible excerpts are selected for each utterance–noise pair. Each excerpt is reused at −5, 0, 5 and 10 dB active-speech SNR. This gives 10 × 3 × 2 × 4 = 240 mixtures for each filter configuration. These are repeated conditions on ten speakers, not 240 independent speech samples. Figure 1 shows the processing and evaluation sequence.

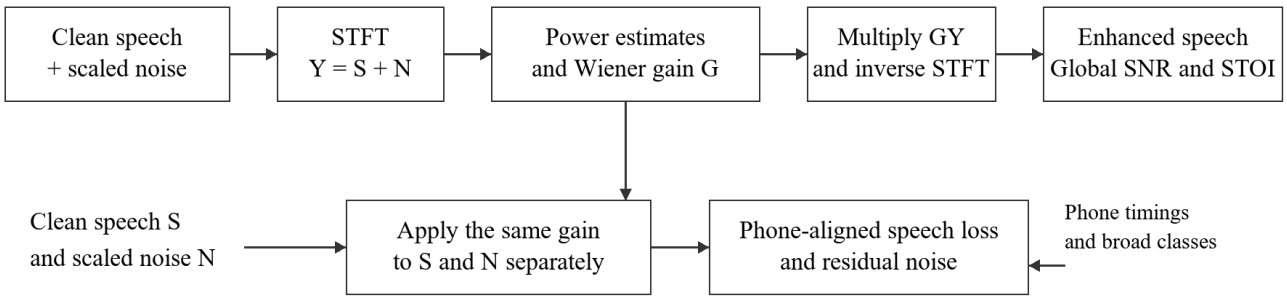


Fig. 1. Signal processing and evaluation.

B. Mixture generation

For each utterance, the code converts the phone boundaries to sample indices and forms an active-speech mask A from the four speech classes. It checks that the labels cover the file without gaps or overlap. A noise excerpt is then selected using the deterministic seed 5305 + 100000i + 1000j + r, where i and j are the speech and noise manifest row indices and r is the repeat number. The excerpt includes a 0.5 s prefix plus the utterance duration.

Let s[n] be clean speech and v[n] the selected, unscaled noise aligned with the utterance. The mixture y[n] and scale α for a requested active-speech SNR are

$$y[n] = s[n] + \alpha v[n], \quad \alpha = \left[\frac{\sum_A s[n]^2}{\sum_A v[n]^2 \cdot 10^{SNR/10}}\right]^{1/2} \tag{1}$$

The same scale applies to the noise-only prefix, where clean speech is zero. Calibration uses only A. Evaluation removes the prefix but includes the original labelled silence, so the whole-utterance input error SNR need not equal the requested active-speech SNR. The noise start sample and scale are saved for every mixture. The code verifies the calibrated SNR to within 10^{-10} dB.

C. STFT configuration

TABLE II. STFT and Wiener filter settings.

Parameter	32 ms baseline	16 ms comparison
Sample rate	16 kHz	16 kHz
Window	Square-root periodic Hann	Same window family
Length / hop	512 / 256 samples	256 / 128 samples
Overlap	50%	50%
FFT length	512	512
Power-smoothing coefficient β	0.8000	0.8944
Noise-only prefix	0.5 s	Same prefix
Minimum amplitude gain	0.05	0.05

The 32 ms baseline provides a practical time–frequency compromise. A half-window of zeros is added before the signal, and the end is padded to a complete frame grid. This permits reconstruction at both ends. The inverse transform uses the same window and hop, and the added padding is removed after synthesis.

D. Power estimation and Wiener gain

For a padded signal x[n], frame length L, hop H and FFT length K, the analysis is

$$X[k,m] = \sum_{l=0}^{L-1} x[mH + l]w[l]\, exp(-j2\pi kl/K) \quad (2)$$

The baseline uses L = K = 512 and H = 256. MATLAB stft and istft provide the transforms [10]. The project code calculates the power estimates, gain and evaluation measures. Reconstruction uses the two-sided STFT, while the spectral metrics use its 257 nonnegative-frequency bins.

First, noise power $P_n[k]$ is estimated as the mean of $|Y[k,m]|^2$ over frames lying wholly within the 0.5 s noise-only prefix. It is held fixed for the utterance. Second, the mixture-power estimate is initialised with $P_n[k]$ and updated at every frame with $\beta = 0.8$:

$$P_y[k,m] = \beta P_y[k,m-1] + (1-\beta)|Y[k,m]|^2 \quad (3)$$

Third, the estimated speech power is the excess of smoothed mixture power above noise power. Negative estimates are set to zero:

$$P_s[k,m] = max(P_y[k,m] - P_n[k], 0) \quad (4)$$

Finally, the gain is the estimated speech-power fraction, limited by an amplitude floor of 0.05:

$$G[k,m] = max(0.05, \frac{P_s[k,m]}{max(P_s[k,m] + P_n[k], \varepsilon)}) \quad (5)$$

The small denominator floor ε is the larger of 10^{-12} times the maximum mixture STFT power and realmin. It prevents division by zero. For uncorrelated speech and noise, minimising $E\{|S - GY|^2\}$ gives the ideal gain $P_s/(P_s + P_n)$. Equations (3)–(5) estimate this ratio from the mixture and prefix. When estimated speech power is small, the gain approaches the floor. When speech power dominates, it approaches one. The 0.05 floor corresponds to about 26 dB maximum attenuation of an individual STFT amplitude.

E. Reconstruction and component separation

The enhanced spectrum is GY. Because G is real and nonnegative, it changes the magnitude while preserving the mixture phase. Weighted overlap-add synthesis multiplies each inverse-transformed frame by the synthesis window and divides the overlapping sum by the sum of squared windows. With the square-root Hann pair and 50% overlap, unity gain reconstructs the input to numerical precision.

To separate speech loss from noise reduction, the code applies the same mixture-derived G to the clean-speech STFT S and scaled-noise STFT N. The reconstructed components are

$$s_G = ISTFT(GS), \quad n_G = ISTFT(GN), \quad \hat{y} = s_G + n_G \quad (6)$$

Equation (6) follows from Y = S + N and the linear synthesis operation when G is fixed. The complete enhancer is nonlinear because its input determines G. Re-estimating the gain from clean speech or noise alone would therefore change the filter being evaluated. Keeping the mixture-derived gain gives a speech component s_G that shows the effect on desired speech and a noise component n_G that shows the remaining interference.

F. Phone alignment

After subtracting the prefix offset, a frame belongs to the phone containing its centre time. Intervals are half-open: the start is included and the end excluded. Each centre is assigned once. Time-domain metrics use the exact phone sample interval after synthesis. Spectral metrics use the assigned frames, whose windows may include neighbouring phones. The fraction of such boundary-crossing frames is retained as a diagnostic.

G. Evaluation measures

For phone interval I, speech retention R_s and noise retention R_n are the output-to-input energy ratios of the separated components:

$$R_s(I) = 10\log_{10}\frac{\sum_I s_G[n]^2}{\sum_I s[n]^2},\quad R_n(I) = 10\log_{10}\frac{\sum_I n_G[n]^2}{\sum_I n[n]^2} \tag{7}$$

Here n[n] denotes the scaled added noise. A value of 0 dB means unchanged interval energy, while −3 dB is approximately half the original energy. Occasional positive time-domain values are retained because overlap-add can redistribute energy across a short phone interval. Silence and segments below 10^{-12} of the utterance speech energy are excluded from speech metrics.

Unweighted mean gain averages G over the assigned frames and all nonnegative-frequency bins. Energy-weighted gain weights each G by the corresponding clean-speech power $|S|^2$. One-sided energy weights are two except at DC and Nyquist, where they are one. The code also averages gain below 1 kHz, from 1 to 4 kHz and from 4 to 8 kHz. High-frequency retention is 10 $\log_{10}$ of the ratio of summed weighted $|GS|^2$ to summed weighted $|S|^2$ in the 4–8 kHz band. It is omitted when the clean band energy is below 10^{-10} of the total frame speech energy.

For candidate waveform z[n], error SNR compares the clean signal with the complete reconstruction error:

$$SNR_e(I,z) = 10\log_{10}\frac{\sum_I s[n]^2}{\sum_I (z[n]-s[n])^2} \tag{8}$$

The improvement ΔSNR_e is $SNR_e(I,\hat{y}) - SNR_e(I,y)$. Applying (8) to the full utterance gives the global measure; applying it to each labelled speech interval gives the segmental measure. The output error includes both modified speech and residual noise. A higher error SNR therefore does not imply that speech remains unchanged.

Log-spectral distortion (LSD) is the mean, over assigned frames, of the root-mean-square difference between log-magnitude spectra:

$$LSD = \frac{1}{M}\sum_{m=1}^{M}\left[\frac{1}{B}\sum_{k=0}^{B-1}(L_{processed}[k,m] - L_{clean}[k,m])^2\right]^{1/2} \tag{9}$$

M is the number of assigned frames and B = 257 is the number of one-sided bins. Log magnitudes use 20 $\log_{10}$ with a floor of 10^{-3} times the maximum clean-STFT magnitude. Mixture LSD compares GY with S; speech-component LSD compares GS with S. The former includes residual noise, while the latter measures gain-induced changes to speech. STOI is computed for each complete utterance using MATLAB Audio Toolbox, rather than for individual short phones.

H. Aggregation and validation

The two repetitions are averaged for each phone, then phones are averaged within each speaker, class, noise and input SNR. Condition summaries give equal weight to each speaker. Pooled summaries first average the 12 noise–SNR conditions within each speaker. Retention summaries average decibel scores rather than taking ratios of pooled sample energies. Standard deviations (SDs) describe variation among the ten speaker means. Paired class comparisons use each speaker's vowel result under the same condition. The 95% percentile intervals resample the ten paired speaker differences 10,000 times with seed 5305. These exploratory intervals are not corrected for multiple comparisons.

Validation checks unity-gain reconstruction, output length, gain limits, independence of the gain from the clean reference, one-sided energy accounting and missing-frame handling. The largest baseline relative reconstruction and decomposition errors are 2.70×10^{-16} and 3.03×10^{-16}. All 549 labels have an assigned baseline frame centre, but 40 of the 506 speech intervals contain no complete 32 ms frame. Missing spectral values are recorded as NaN with valid counts, rather than replaced by zero.

I. Reproduction

Run main from the project folder in MATLAB R2025b with Signal Processing Toolbox; Audio Toolbox is required for STOI. Run validateDesign for numerical checks and compareWindows for the paired experiment. The data, labels, manifests and saved result tables are included in the repository linked on the first page.

IV. Results

A. Whole-utterance performance

Global error SNR improved in all 12 noise–SNR condition means, averaging 6.18 dB, but mean STOI fell from 0.8432 to 0.8385 (−0.0047). In cafeteria noise at −5 dB, error SNR improved by 5.29 dB while STOI fell by 0.0389. Lower reconstruction error therefore did not consistently improve predicted intelligibility. Table III reports SNR changes as mean ± SD across ten speakers and STOI as means.

TABLE III. Whole-utterance SNR and STOI.

Noise	Active SNR (dB)	Δ error SNR (dB)	Noisy STOI	Enhanced STOI
Kitchen	-5	7.54 ± 1.94	0.8955	0.9015
Kitchen	0	7.42 ± 1.89	0.9426	0.9450
Kitchen	5	7.24 ± 1.83	0.9721	0.9728
Kitchen	10	6.97 ± 1.78	0.9881	0.9882
Cafeteria	-5	5.29 ± 0.58	0.5753	0.5364
Cafeteria	0	4.69 ± 0.58	0.7015	0.6824
Cafeteria	5	4.09 ± 0.63	0.8154	0.8086
Cafeteria	10	3.51 ± 0.68	0.8995	0.8991
Traffic	-5	7.88 ± 2.28	0.6983	0.6981
Traffic	0	7.18 ± 1.94	0.8043	0.8042
Traffic	5	6.49 ± 1.75	0.8863	0.8859
Traffic	10	5.90 ± 1.73	0.9398	0.9402

Mean error-SNR gains across input SNRs were 7.29 dB for kitchen, 4.39 dB for cafeteria and 6.86 dB for traffic noise. These values apply to the selected recordings. Stationarity was not measured, so the experiment does not establish which noise properties caused the differences. Across the full utterances, mean speech retention was −0.66 dB and mean noise retention was −6.89 dB.

B. Applied gain by phonetic class

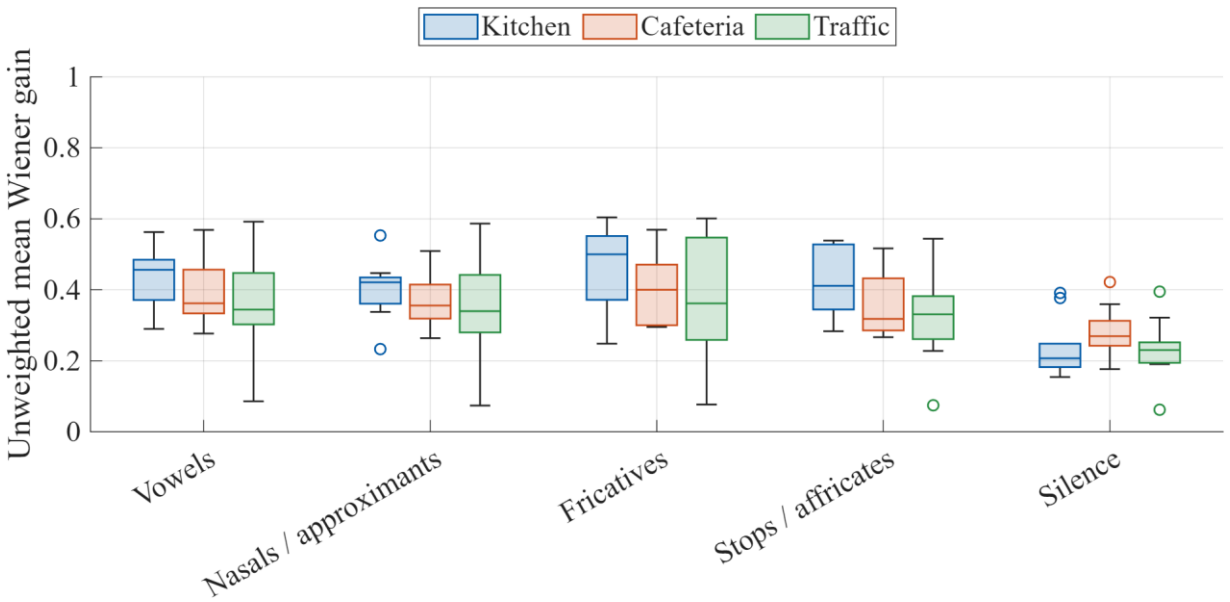


Fig. 2. Mean Wiener gain by class at 0 dB input SNR.

Boxes in Fig. 2 summarise ten speaker means: lines mark medians, boxes span the interquartile range (IQR), whiskers end at the most extreme values within 1.5 IQRs of the box, and circles mark outliers. Silence has the lowest mean gain: 0.235, 0.277 and 0.234 in kitchen, cafeteria and traffic noise. Fricative means are 0.462, 0.403 and 0.370, above the vowel means of 0.439, 0.384 and 0.361. Higher unweighted gain can therefore accompany greater speech loss.

C. Speech retention across noise and SNR

Mean speech retention moves towards 0 dB as input SNR increases in all four classes (Table IV). Values average ten speakers; more negative values indicate greater speech attenuation. SNR denotes the active-speech input setting. Class differences remain at the highest SNR. At 10 dB in kitchen noise, vowel retention is −0.04 dB and fricative retention is −0.61 dB. In cafeteria noise at the same SNR, they are −0.52 and −1.36 dB.

TABLE IV. Speech retention by noise and input SNR (dB).

Noise	SNR (dB)	Vowels	Nasals / approximants	Fricatives	Stops / affricates
Kitchen	-5	-0.24	-0.31	-2.02	-1.44
Kitchen	0	-0.12	-0.16	-1.37	-0.91
Kitchen	5	-0.07	-0.09	-0.93	-0.58
Kitchen	10	-0.04	-0.05	-0.61	-0.38
Cafeteria	-5	-4.74	-5.28	-5.85	-6.28
Cafeteria	0	-2.58	-2.91	-3.85	-4.07
Cafeteria	5	-1.23	-1.45	-2.34	-2.42
Cafeteria	10	-0.52	-0.68	-1.36	-1.38
Traffic	-5	-2.87	-3.45	-5.13	-5.46
Traffic	0	-1.48	-1.88	-3.47	-3.55
Traffic	5	-0.72	-1.01	-2.19	-2.27
Traffic	10	-0.32	-0.50	-1.31	-1.40

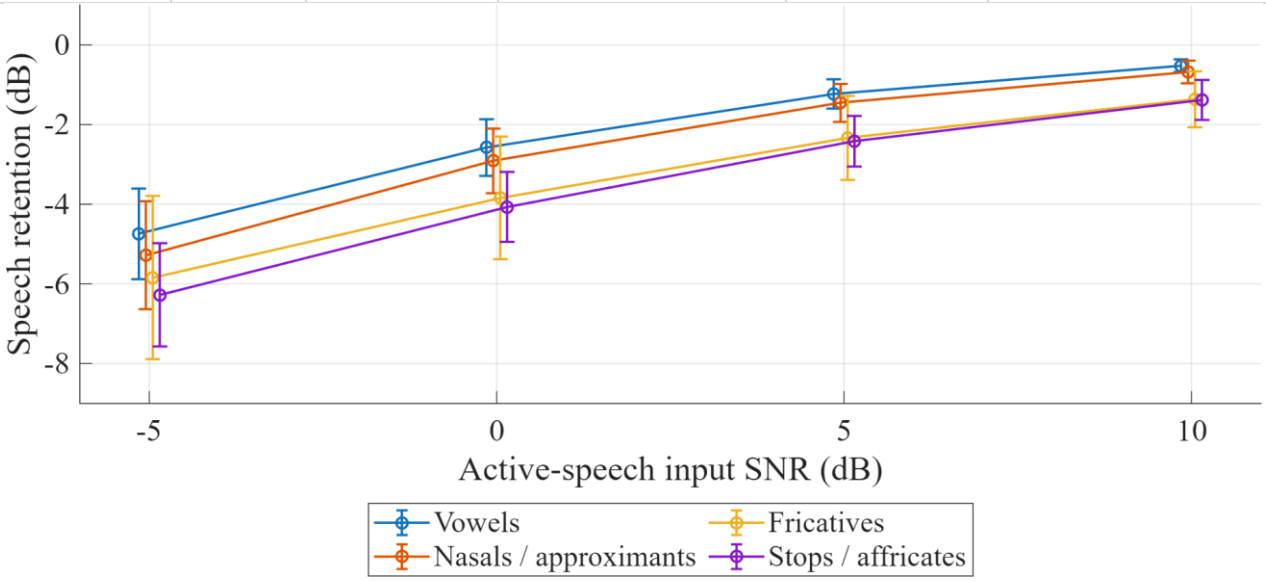


Fig. 3. Speech retention in cafeteria noise.

In Fig. 3, points and error bars show the mean and SD across ten speakers; horizontal offsets keep the bars visible. At −5 dB, vowel retention is −4.74 dB, compared with −5.85 dB for fricatives and −6.28 dB for stops and affricates. The paired differences from vowels are −1.11 dB (95% interval −2.08 to −0.25) and −1.54 dB (−2.29 to −0.80). Segmental error SNR still improves in every class–noise–SNR mean. At −5 dB, improvements averaged over noise types are 8.17 dB for vowels, 8.66 dB for nasals/approximants, 9.50 dB for fricatives and 9.57 dB for stops/affricates. The total reconstruction error thus falls despite the loss of desired speech.

D. Component and spectral measurements

TABLE V. Component retention and weighted gain.

Class	Speech retention (dB)	Noise retention (dB)	Weighted gain
Vowels	-1.24 ± 0.31	-8.67 ± 1.88	0.875
Nasals / approximants	-1.48 ± 0.51	-8.84 ± 2.13	0.851
Fricatives	-2.53 ± 1.14	-9.02 ± 1.90	0.749
Stops / affricates	-2.51 ± 0.85	-9.01 ± 2.11	0.742

Table V reports pooled retention as mean ± SD of ten speaker means. Relative to vowels, fricatives and stops/affricates have 1.29 and 1.27 dB lower speech retention, but only 0.35 and 0.34 dB lower noise retention. Energy weighting reverses the gain ranking: 0.749 for fricatives versus 0.875 for vowels. Unweighted averaging hides the gain applied where speech energy is concentrated.

TABLE VI. Frequency-band gain and spectral measures.

Class	Gain <1 kHz	Gain 1–4 kHz	Gain 4–8 kHz	4–8 kHz retention (dB)	Mixture LSD (dB)	Speech LSD (dB)
Vowels	0.615	0.496	0.374	-6.91	6.45	4.43
Nasals / approximants	0.610	0.457	0.353	-8.06	6.43	3.63
Fricatives	0.536	0.446	0.461	-6.08	6.82	3.74
Stops / affricates	0.562	0.448	0.368	-7.55	6.72	3.22

Fricatives have the highest pooled 4–8 kHz gain (0.461) and retention (Table VI). Their retention is −6.08 dB, versus −6.91 dB for vowels. Greater full-band loss therefore does not imply stronger suppression in every frequency band.

Mixture LSD is higher for fricatives than vowels (6.82 versus 6.45 dB), but speech-component LSD is lower (3.74 versus 4.43 dB). Only mixture LSD includes residual noise. Both log-spectral measures emphasise low-energy bins more than energy ratios do. Figure 4 shows the example from Section IV-F; brighter regions indicate larger gains.

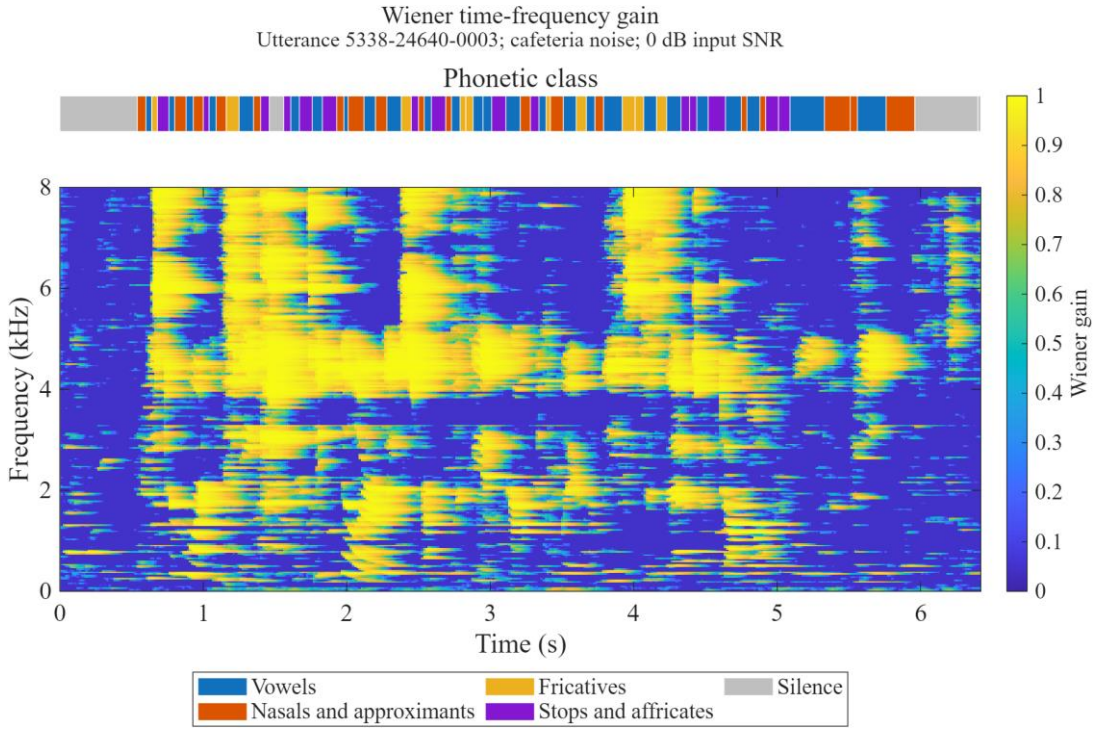


Fig. 4. Wiener gain and phone alignment.

E. Effects of noise and input SNR

Table VII compares fricatives with vowels for each noise and input SNR. Every contrast pairs the same ten speakers and gives them equal weight. It shows which pooled differences in Tables V and VI hold across conditions. All values are in dB. A negative retention difference means that fricatives lose a larger proportion of speech energy; a positive LSD difference means greater fricative distortion.

TABLE VII. Global SNR improvement and fricative–vowel differences (dB).

Noise	SNR (dB)	Global ΔSNR_e	Speech retention F − V	Mixture LSD F − V	Speech LSD F − V	4–8 kHz retention F − V
Kitchen	-5	7.54	-1.78	+0.67	+0.13	+1.65
Kitchen	0	7.42	-1.25	+0.55	+0.14	+1.07
Kitchen	5	7.24	-0.86	+0.42	+0.11	+0.34
Kitchen	10	6.97	-0.57	+0.30	+0.02	-0.20
Cafeteria	-5	5.29	-1.11	+0.43	-1.52	+1.85
Cafeteria	0	4.69	-1.27	+0.42	-1.32	+1.39
Cafeteria	5	4.09	-1.10	+0.57	-0.96	+0.74
Cafeteria	10	3.51	-0.84	+0.74	-0.61	+0.18
Traffic	-5	7.88	-2.25	-0.29	-1.65	+1.58
Traffic	0	7.18	-1.99	-0.05	-1.30	+1.01
Traffic	5	6.49	-1.47	+0.29	-0.86	+0.47
Traffic	10	5.90	-0.99	+0.49	-0.53	-0.17

Fricatives have lower full-band speech retention than vowels in all 12 conditions. The largest gap is −2.25 dB in traffic noise at −5 dB, where global error SNR improves by 7.88 dB. Cafeteria noise at −5 dB gives a smaller global improvement of 5.29 dB and a smaller class gap of −1.11 dB. Thus, the stronger global improvement in the traffic condition accompanies a larger difference in speech loss between the two classes.

Spectral distortion does not follow the retention ranking. Mixture LSD is higher for fricatives in ten conditions, but lower in traffic noise at −5 and 0 dB. Speech-component LSD is higher for fricatives in kitchen noise and lower in cafeteria and traffic noise at every SNR. At −5 dB in traffic noise, fricatives lose a larger proportion of energy than vowels yet have lower distortion on both LSD measures. Energy loss and log-spectral distance therefore describe different changes to the signal.

Fricatives retain a larger proportion of their 4–8 kHz energy in ten conditions. At 10 dB, kitchen and traffic noise reverse this ordering, with fricative-minus-vowel differences of −0.20 and −0.17 dB. These are averages across speakers; individuals may show a different ranking. The pooled high-frequency result should therefore be read alongside the values for each condition.

The metric definitions explain these different rankings. Global output error is $(s_G - s) + n_G$, so its energy includes speech distortion, residual noise and their cross-term. Removing noise can reduce the total error even when speech energy falls. Retention measures the energy change, while LSD compares log magnitudes across bins, including regions with little clean speech energy. Within each speaker, class summaries average the decibel scores of individual phones, whereas whole-utterance SNR uses total signal and error energies. The retention gap identifies unequal attenuation, but it does not fully describe the output.

F. Speech example

The example uses the first manifest utterance, 5338-24640-0003, with cafeteria noise at 0 dB active-speech SNR and repeat 1. Selection follows manifest order and does not depend on the output score. Whole-utterance error SNR rises from −0.82 to 3.97 dB, an improvement of 4.79 dB. STOI rises from 0.6936 to 0.7156. Both measures improve for this utterance, unlike the negative mean STOI change across all conditions. The three spectrograms in Fig. 5 share the clean-peak reference and a −70 to 0 dB colour scale. Time starts at the original utterance, excluding the calibration prefix.

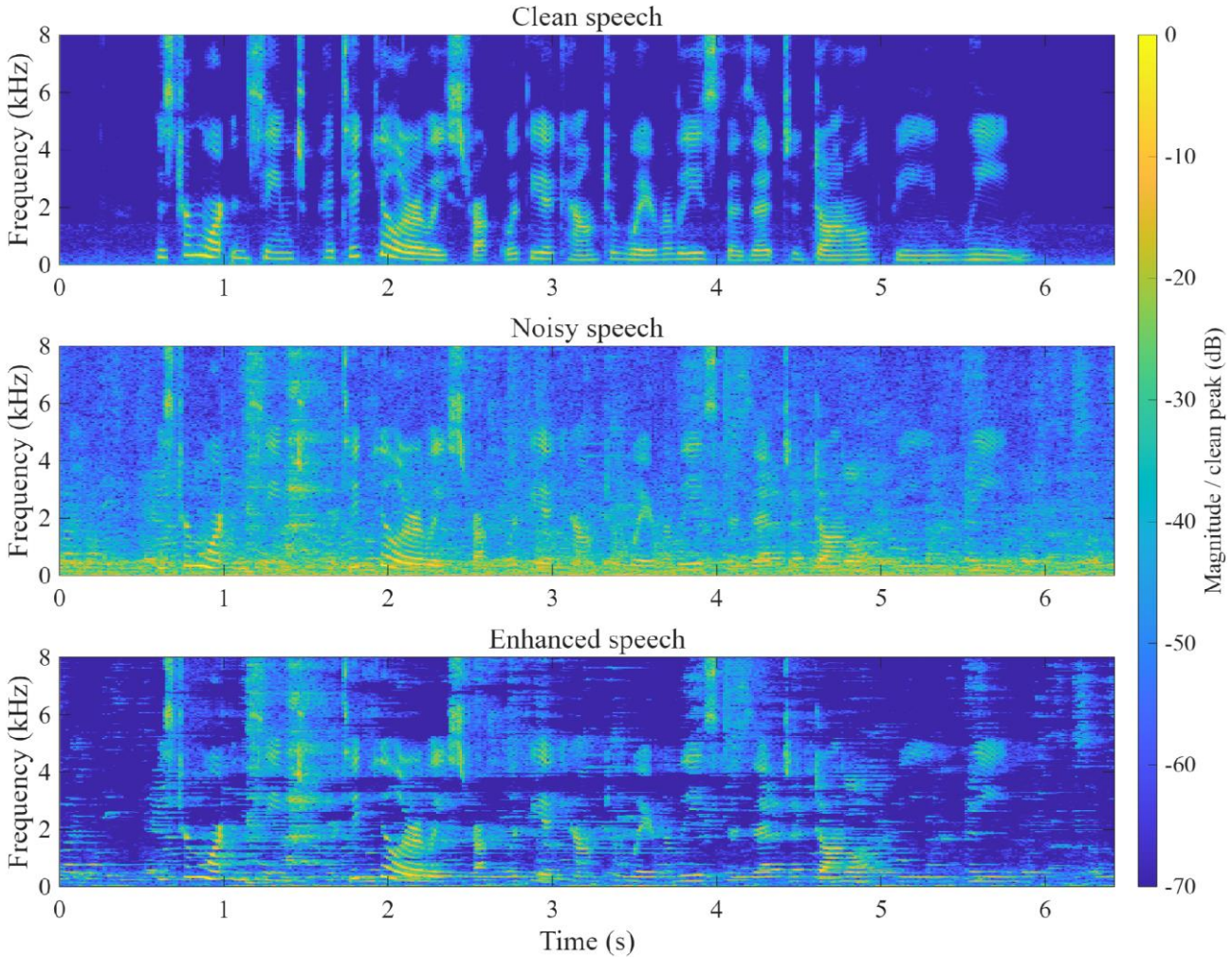


Fig. 5. Clean, noisy and enhanced speech spectrograms.

In Fig. 5, noise fills many of the low-energy regions between speech bands. Filtering lowers much of this background, while prominent bands remain visible. The gain map in Fig. 4 shows that suppression changes within individual phones as well as between them. A bright spectral region may therefore remain beside a strongly attenuated region of the same sound. Some visible energy can still be residual noise, so the spectrograms are interpreted together with the separated components.

For this utterance, whole-speech retention is −1.27 dB and noise retention is −5.51 dB. The output error depends on speech distortion, residual noise and their interaction, as described in Section IV-E. The clean, noisy and enhanced audio files share one playback scale, preserving their relative levels. All metrics were calculated before playback scaling.

G. Local spectra and gain

Figures 6 and 7 show the vowel and fricative with the most assigned frames in this utterance. Averaging these frames gives a local spectrum for each sound, although the selection excludes the shortest consonants. The upper panels compare clean, noisy and enhanced spectra; the lower panels show mean amplitude gain over the same frames. Each spectral panel is normalised to its own clean-spectrum peak, so absolute levels cannot be compared between phones.

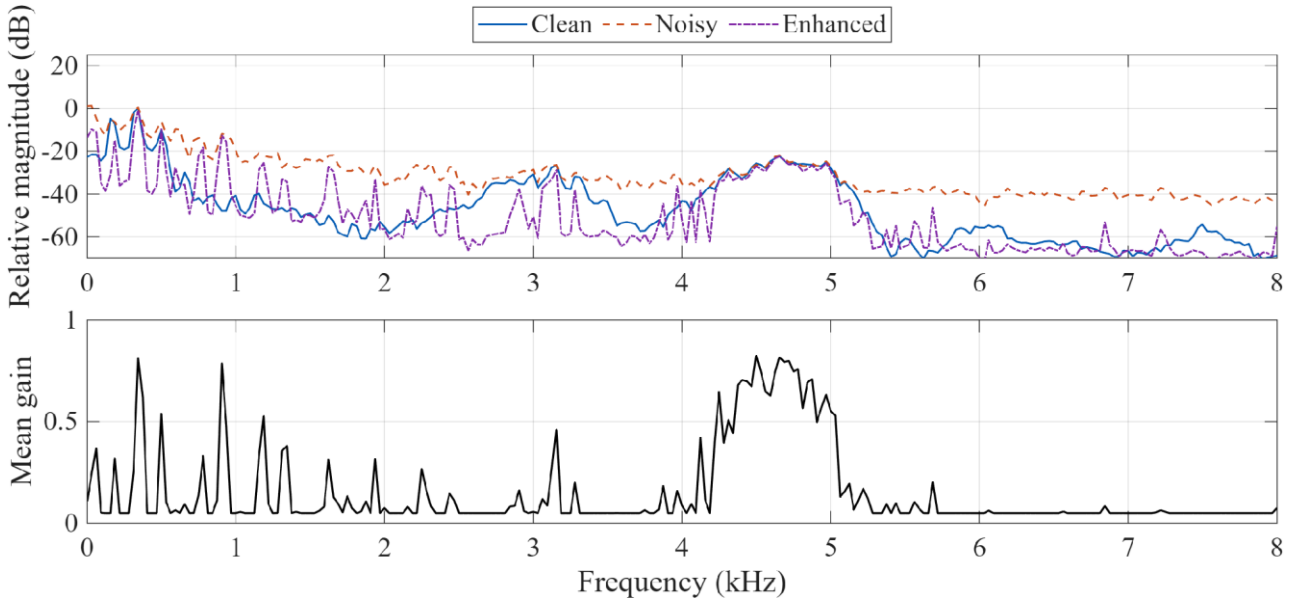


Fig. 6. Vowel IY spectrum and gain, 5.09–5.33 s.

The IY interval has speech retention of −5.27 dB and a clean-energy-weighted gain of 0.497. The gain curve shows suppression in frequency regions containing vowel energy. Residual noise remains in the noisy and enhanced spectra, so the separated speech component is needed to quantify the vowel energy lost.

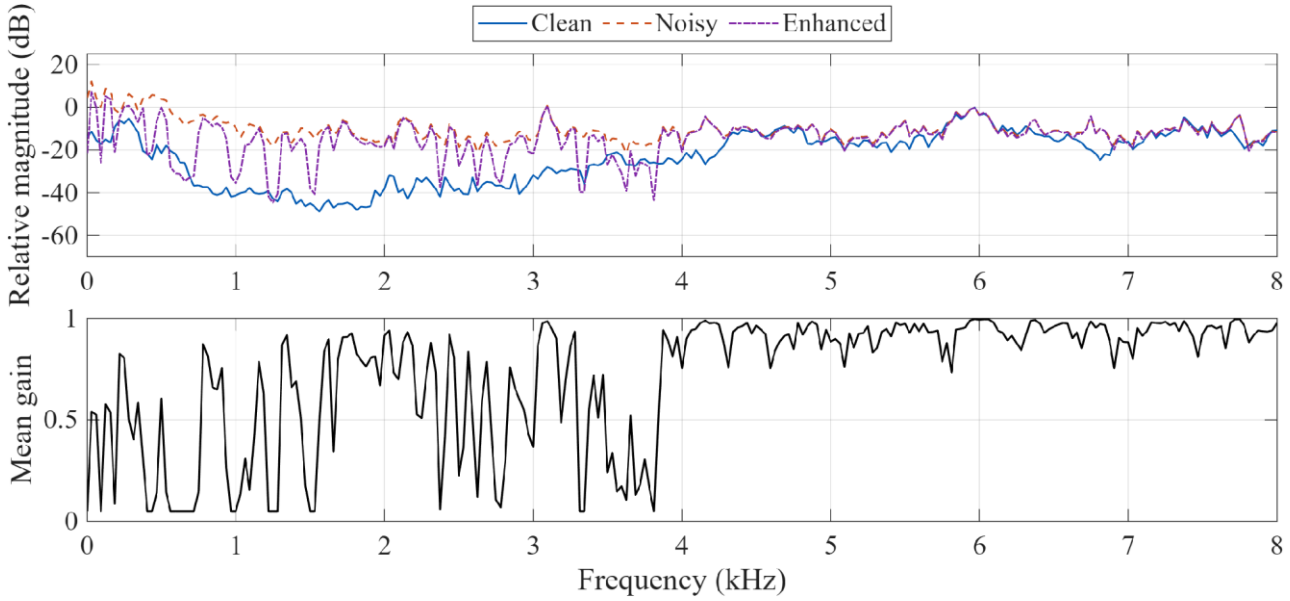


Fig. 7. Fricative S spectrum and gain, 1.16–1.25 s.

The S interval has speech retention of −0.33 dB and a weighted gain of 0.909. Much of its high-frequency structure remains after filtering, and it retains a larger proportion of its speech energy than the selected vowel. The class averages have the opposite ordering. This difference shows why local spectral energy and gain must be considered when interpreting an individual phone.

V. Window-Length Experiment

Forty speech intervals contain no complete 32 ms frame, and many assigned frames extend into neighbouring phones. To reduce this overlap, the comparison uses a 16 ms window with an 8 ms hop. Speech, labels, noise excerpts and scaling remain unchanged. The 512-point FFT keeps the bin spacing at 31.25 Hz, but zero padding cannot recover the frequency resolution lost by shortening the window [11].

The shorter hop also requires a change in power smoothing. The decay time constant is $-H/(f_s \ln \beta)$, so halving H requires changing β from 0.8 to $\sqrt{0.8} = 0.8944$ to keep the same decay time. Leaving β unchanged would make adaptation twice as fast. All 240 mixtures are processed with the shorter window. Table VIII pools noise and SNR conditions within each speaker, then compares the configurations. The 95% intervals resample the ten paired speaker differences as described in Section III-H.

TABLE VIII. Comparison of 32 ms and 16 ms windows.

Measurement	32 ms	16 ms	16 − 32 ms	95% interval for change
Global Δ error SNR (dB)	6.182	6.245	+0.063	[-0.043, +0.166]
STOI change	-0.0047	-0.0140	-0.0093	[-0.0110, -0.0077]
Global noise retention (dB)	-6.889	-7.251	-0.362	[-0.491, -0.249]
Vowel speech retention (dB)	-1.244	-1.721	-0.477	[-0.561, -0.396]
Nasal/approximant retention (dB)	-1.481	-2.090	-0.609	[-0.768, -0.474]
Fricative speech retention (dB)	-2.535	-3.128	-0.593	[-0.711, -0.486]
Stop/affricate retention (dB)	-2.512	-3.140	-0.628	[-0.767, -0.484]

The fraction of frames crossing phone boundaries falls from 42.2% to 19.9% for vowels, 46.2% to 21.2% for nasals/approximants, 45.6% to 20.7% for fricatives and 42.7% to 19.4% for stops/affricates. Despite this improvement in timing, mean speech retention decreases by 0.477–0.628 dB. After averaging the 12 conditions, each speaker has lower retention in every speech class. Retention is measured on exact time-domain intervals, so this loss is more than a change in which frames receive each label.

Across the four classes, clean-energy-weighted gain falls by 0.029–0.041, while unweighted mean gain rises slightly, by 0.002–0.005. The weighted result points to stronger attenuation of the regions carrying speech energy, which agrees with the loss measured after reconstruction. These averages come from different STFT windows and therefore do not compare identical spectral estimates. They describe the overall gain pattern rather than isolating a single cause of the extra attenuation.

The pooled change in global error-SNR improvement is +0.063 dB, with a 95% interval that includes zero. The condition means are positive for all three noises at −5 dB but negative for all three at 10 dB. In cafeteria noise at −5 dB, the shorter window lowers noise retention by 1.04 dB and whole-speech retention by 0.93 dB. STOI is lower in every noise–SNR condition. The small pooled error-SNR benefit therefore comes with greater speech loss and a lower intelligibility prediction.

Shortening the window reduces overlap with neighbouring phones but broadens its spectral response, reducing frequency resolution. Both changes can affect the power estimates and gain. Matching the smoothing time constant controls the estimator's decay time, but the spectral estimates and correlations between frames still differ. This comparison cannot isolate their contributions to the extra speech suppression. For these settings, the 32 ms baseline retains more speech, while the 16 ms window produces fewer frames that cross phone boundaries.

VI. Discussion

A. Contribution and relation to previous work

The separate speech and noise measurements show how a lower global error can conceal unequal speech loss. Monir et al. [3] also found differences among phonemes, including benefits for sibilants. The preserved S example is consistent with that observation, although the systems, metrics and acoustic conditions differ. Kim and Loizou [2] found a relatively small intelligibility cost from attenuation under their test conditions. Energy loss and STOI should therefore be interpreted separately from human recognition. The window comparison extends the baseline by showing that fewer boundary-crossing frames do not, by themselves, improve speech retention.

B. Limitations and future work

The ten speakers contribute one utterance each, so the sample has limited coverage. Class differences may partly reflect duration, energy, speaking style or context, even with equal speaker weights. Broad classes combine different acoustic types, and stop intervals include more than release bursts. Label checks verified timing consistency rather than the accuracy of every boundary. More utterances and manual boundary checks would help test whether the findings generalise.

Only one recording represents each noise environment. Later noise may differ from the fixed prefix estimate, but this mismatch was not measured. Filter parameters were not exhaustively tuned. LSD depends on its magnitude floor, weak components affect band-energy ratios, and the 8 kHz limit excludes information discussed in [5]. Noise tracking or a different gain floor could be tested on separate material, followed by listening tests of recognition.

VII. Conclusion

For these 240 mixtures, the Wiener filter improves mean global error SNR by 6.18 dB while attenuating speech classes unequally. Fricatives retain a smaller proportion of their full-band speech energy than vowels in every noise–SNR condition, although spectral and high-frequency rankings vary. Global error SNR therefore misses part of the effect on speech. Shorter windows reduce boundary overlap but increase attenuation without a consistent global benefit. The experiment answers the research question for signal-level preservation under the tested conditions; perceptual effects remain unmeasured.

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